Programming Fundamentals

VARIABLES & SCOPE

Question 1: What is a variable in C/C++?

A named location in memory used to store data that can be modified during program execution.

Question 2: Why do we use variables in programs?

To dynamically store user inputs, manage shifting program states, and reuse computational values without hardcoding them.

Question 3: What is the difference between a variable name and its value?

The **name** is the identifier label used to reference the memory location, while the **value** is the actual content stored inside that location.

Question 4: Which of the following is a valid variable name?

b) total_marks. (Others are invalid: 2num starts with a number, for is a keyword, and my marks contains a space).

Question 5: Why is Age different from age?

C/C++ is strictly case-sensitive. Capital and lowercase letters are processed as completely distinct characters by the compiler.

Question 6: Can a variable name contain spaces? Why?

No. Spaces act as code separators. Adding a space splits the identifier into separate tokens, causing a compilation syntax error.

Question 7: Can a variable name start with an underscore (_) ?

Yes, it is syntactically legal, but best avoided to prevent naming conflicts with internal system libraries and macro definitions.

Question 8: What happens if two variables have the same name in the same scope?

It triggers a "redefinition error" compilation failure because the compiler cannot resolve the naming ambiguity.

Question 9: Write three meaningful variable names for storing student information.

`roll_number, student_gpa, and email_address.`

Question 10: Which naming convention is better and why: studentMarks or sm?

studentMarks is better because descriptive names make the source code readable, self-documenting, and easier to maintain.

DATA TYPES

Question 11: What is a data type?

An attribute that tells the compiler how much memory space to allocate, what format to store the data in, and what operations are valid.

Question 12: Which data type is used to store whole numbers?

Int data type

Question 13: Which data type is used to store decimal values?

`float or double.`

Question 14: What is the difference between float and double?

float takes 4 bytes of memory , double takes 8 bytes

Question 15: Which data type stores a single character?

The char data type.

Question 16: Which data type stores True/False values?

The bool data type.

Question 17: What data type is suitable for storing a person's name?

The std::string data type.

Question 18: Predict the output: int age=18; cout<<age;

Output: 18

Question 19: Identify the data type: 25, 3.14, 'A', true.

25 is an int, 3.14 is a double, 'A' is a char, and true is a bool.

Question 20: What happens if you store a decimal number in an int variable?

The decimal portion is completely discarded (truncated) without any rounding (e.g., 5.99 becomes 5).

INPUT & OUTPUT STREAM OPERATIONS

Question 21: What is the purpose of cin?

It is the standard input stream object used to read input data directly from the keyboard.

Question 22: What is the purpose of cout?

It is the standard output stream object used to display text or variable values on the terminal screen.

Question 23: Which symbol is used with cin?

The stream extraction operator: >>.

Question 24: Which symbol is used with cout?

The stream insertion operator: <<.

Question 25: Explain: cin >> age;

Pauses execution to capture user input from the keyboard, parses it, and copies the value into the variable age.

Question 26: Explain: cout << age;

Fetches the data stored in the variable age and sends it to the console stream to be displayed.

Question 27: What is the purpose of endl?

It inserts a newline character ('\n') into the output stream and forces an immediate flush of the stream buffer.

Question 28: Write a statement to input a student's marks.

```
cin >> student_marks;
```

Question 29: Write a statement to display 'Welcome to C++'.

```
cout << "Welcome to C++";
```

Question 30: Predict the output: cout << 'Programming';

It causes a compilation warning/error or prints an unexpected numeric value. Single quotes can only hold a single character literal; words must use double quotes ("Programming").

OPERATORS & EXPRESSIONS

Question 31: What is an operator?

A symbol that tells the compiler to perform specific mathematical, relational, or logical computations on operands.

Question 32: Which operator is used for addition?

The plus operator: +.

Question 33: What is the result of 10 % 3?

Result:1. (The modulo operator computes integer remainder: 10 divided by 3 leaves a remainder of 1).

Question 34: What is the result of 8 % 2?

Result:0. (Since 8 is perfectly divisible by 2 with no remainder).

Question 35: Which operator checks equality?

The comparison operator: ==.

Question 36: What is the difference between = and == ?

= is the assignment operator (stores right-side value into left-side variable). == is the equality operator (checks if two values are equal).

Question 37: What is the result of 5 > 3?

Result:true (evaluates numerically to integer 1).

Question 38: What does && represent?

The logical **AND** operator. It returns true if and only if all evaluated conditions are true.

Question 39: What does || represent?

The logical **OR** operator. It returns true if at least one of the conditions is true.

Question 40: What does ! represent?

The logical **NOT** operator. It inverts a boolean value (turns true to false, or false to true).

PROGRAMMING LOGIC & CALCULATIONS

Question 41: Write a program to print 'Hello World'.

```
#include <iostream>
using namespace std;

int main() {
    cout << "Hello World" << endl;
    return 0;
}
```

Question 42: Write a program to input two numbers and display their sum.

```
#include <iostream>
using namespace std;

int main() {
    double num1, num2;
    cin >> num1 >> num2;
    cout << "Sum: " << (num1 + num2) << endl;
    return 0;
}
```

Question 43: Write a program to calculate the area of a rectangle.

```
#include <iostream>
using namespace std;

int main() {
    double length, width;
    cin >> length >> width;
    cout << "Area: " << (length * width) << endl;
    return 0;
}
```

Question 44: What formula is used to calculate Simple Interest?

```
Simple Interest = (Principal * Rate * Time) / 100
```

Question 45: Write a program to calculate Simple Interest.

```
#include <iostream>
using namespace std;

int main() {
    double p, r, t;
    cin >> p >> r >> t;
    cout << "Interest: " << (p * r * t) / 100.0 << endl;
    return 0;
}
```

Question 46: Why is a temporary variable used while swapping two numbers?

Direct copying overwrites data. If you write a = b; without saving the original value of value `a` somewhere first, its is overwritten and lost forever.

Question 47: Swap the values: a=10, b=20.

```
int temp = a;
a = b;
b = temp;
```

Question 48: What formula is used to convert Celsius into Fahrenheit?

```
Fahrenheit = (Celsius * 1.8) + 32
```

Question 49: What is the Fahrenheit value of 0°C?

32°F

Question 50: Why can integer division give incorrect results in the Celsius-to-Fahrenheit program?

If you use 9 / 5 in code, the compiler treats it as an integer operation, drops the remainder, and truncates it to 1. This completely ruins the conversion calculation. You must use 9.0 / 5.0 to enforce accurate floating-point math.